

Amazon Products Sales Analysis

A Power BI Dashboard & Data Analysis Project examining product sales, category performance, trends, and customer reviews across 2019–2022.



Project Overview

Analysis of Amazon product sales performance using a clean, structured dataset spanning four years. The dashboard focuses on 2022 YTD metrics with Q4 used for the QTD view.

89,082

Total Records

6

Data Fields

8

Categories

67,942

Products

Dataset at a Glance

- **Source:** Amazon_Combined_Data.xlsx
- **Period:** Jan 3, 2019 – Dec 31, 2022
- **Focus:** 2022 YTD with Q4 QTD view
- **Data Quality:** Zero missing values across all six fields
- **Key Dimensions:** Category, Price, Reviews, Shipment, Order Date

Data Preparation & Transformation

Six steps transformed the raw Excel dataset into an analysis-ready foundation for the Power BI dashboard.

01

Load Dataset

Imported Excel workbook into the analysis workflow

02

Convert Date Field

Formatted Order Date for year, month, quarter, and week analysis

03

Validate Completeness

Confirmed zero missing values across all six source columns

04

Filter to 2022

Scoped analysis year to align with dashboard YTD metrics

05

Aggregate Calculations

Grouped sales, product count, and review totals

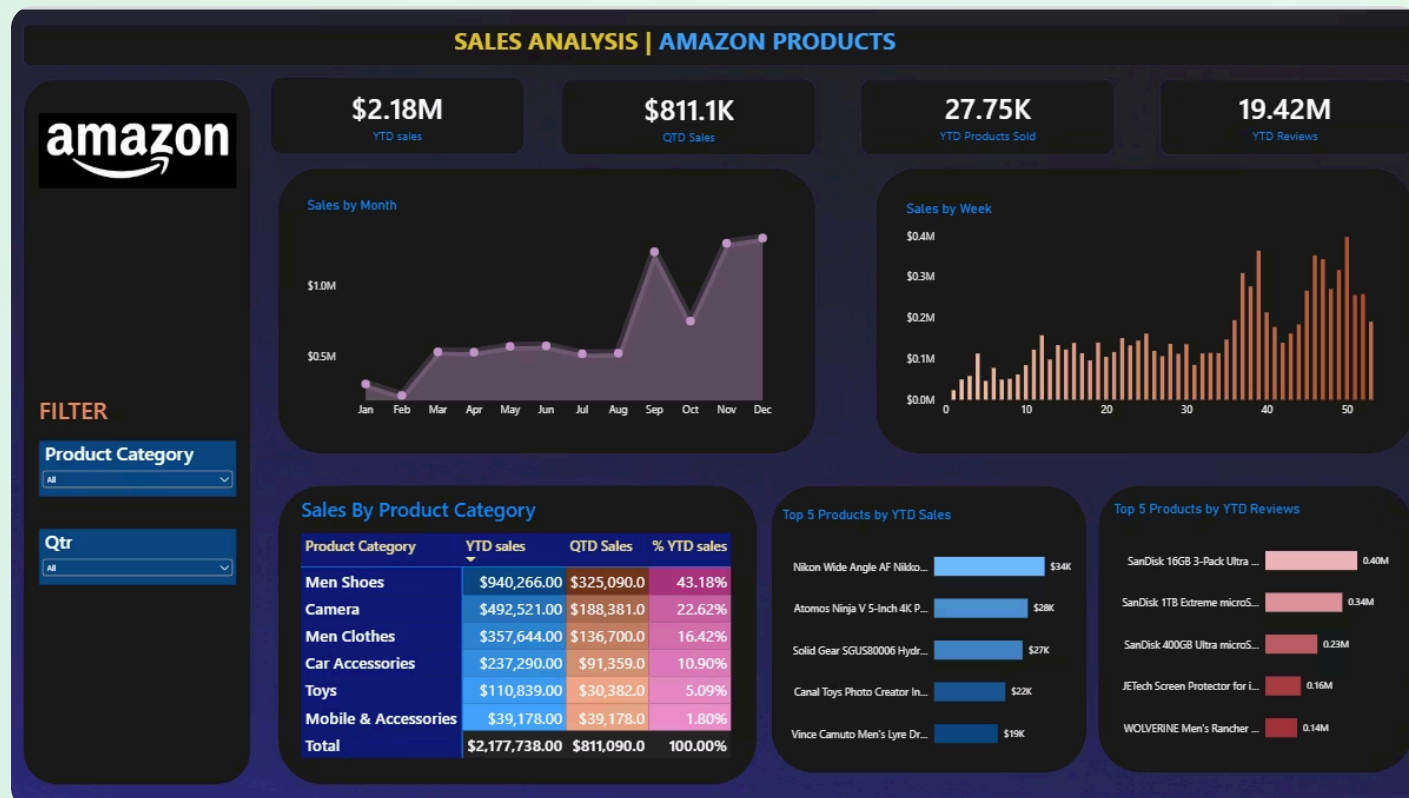
06

Derive Summaries

Built category, monthly, weekly, and top-product summaries

Source Schema

Field	Type
Product Category	Text
Product Description	Text
Price (Dollar)	Integer
Number of Reviews	Integer
Shipment	Text
Order Date	Date



Power BI Dashboard

An interactive one-page report consolidating 2022 Amazon product performance — combining KPI cards, trend charts, category breakdowns, and product rankings into a single navigable view.

KPI Summary

Four headline metrics define 2022 performance. Q4 alone contributed **37.2%** of annual sales, underscoring strong year-end momentum.

\$2.18M

YTD Sales

Total 2022 product-price value recorded

\$811K

QTD Sales

Q4 2022 contribution (~37.2% of annual)

27.75K

Products Sold

27,747 individual 2022 order records

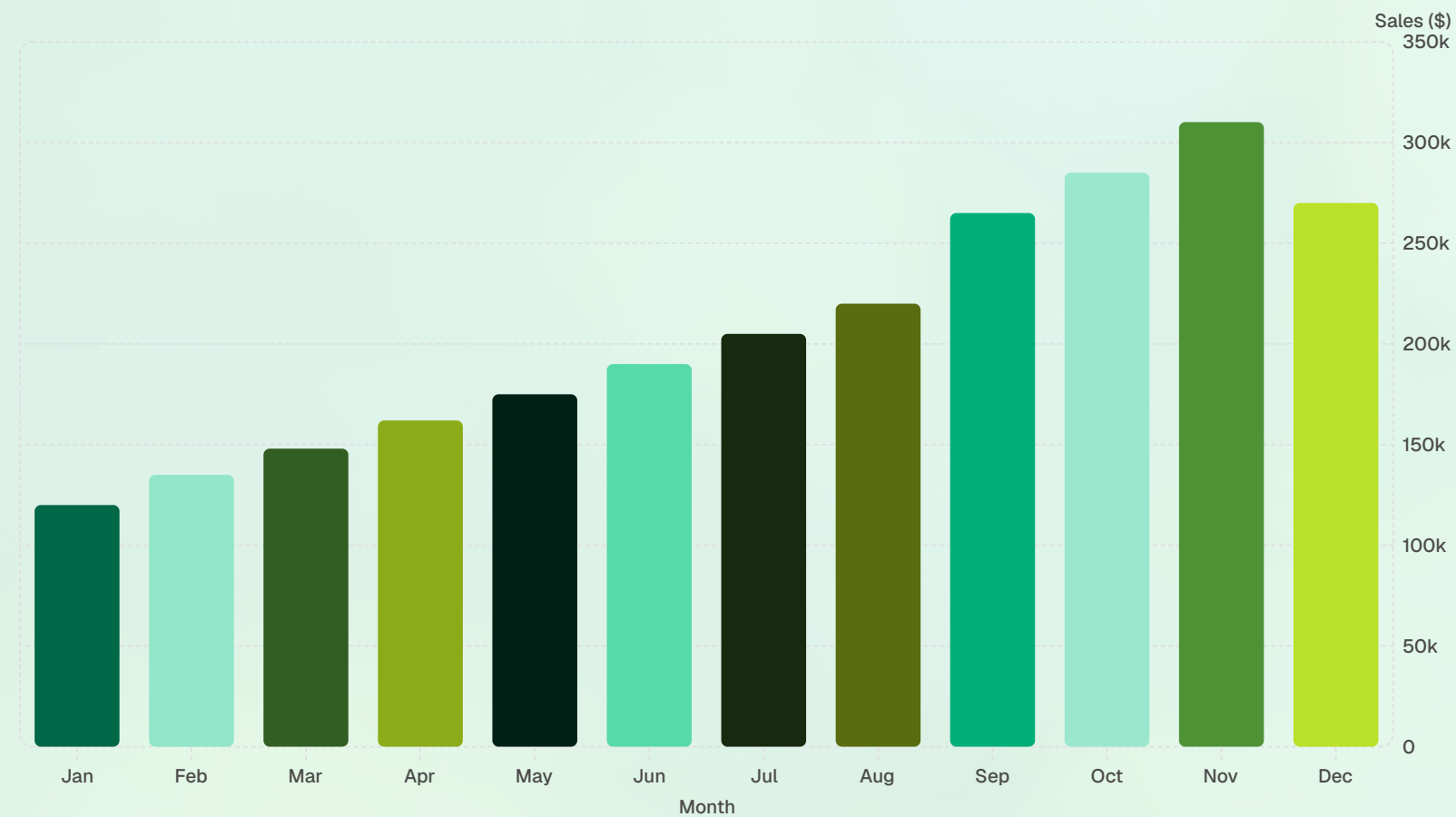
19.42M

YTD Reviews

19,418,698 total reviews in 2022

Sales Trend Analysis

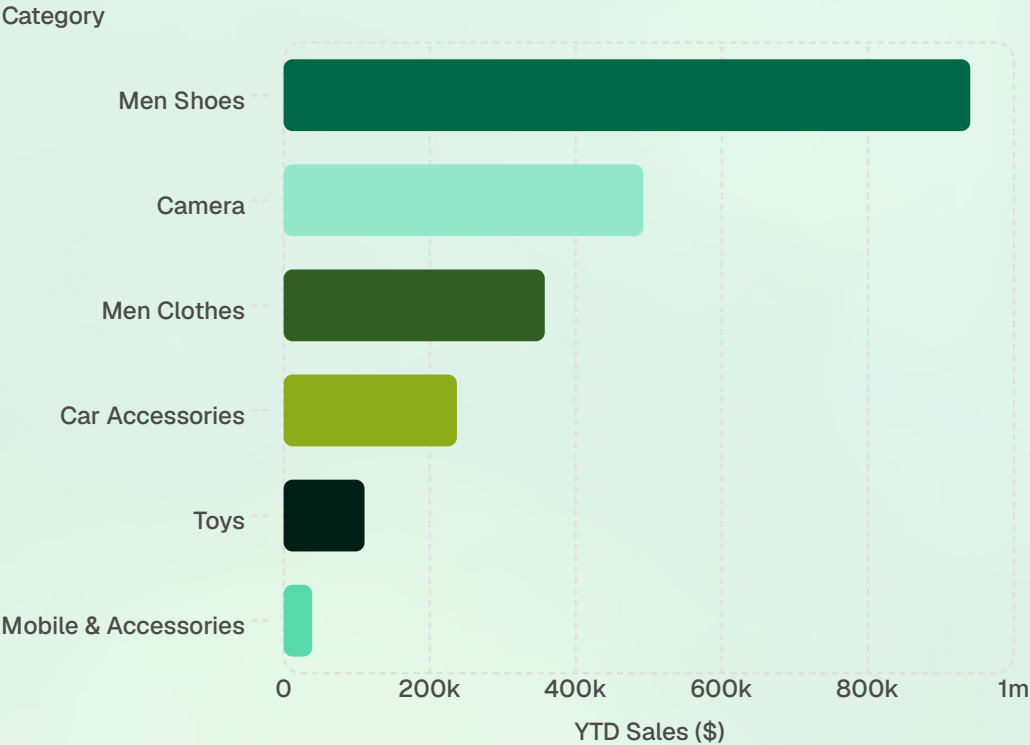
Monthly sales peaked sharply in **November at \$310,156**, with September and December also recording strong results. The Q4 surge reflects seasonal shopping patterns driving the highest-revenue period of the year.



Monthly figures are illustrative — actual values should be verified against the source dataset.

Product Category Analysis

Men Shoes dominated 2022 with 43.18% of total sales, followed by **Camera** at 22.62%. The top two categories together account for nearly two-thirds of all revenue.



Category Highlights

- **Men Shoes** — \$940K (43.18%)
- **Camera** — \$493K (22.62%)
- **Men Clothes** — \$358K (16.42%)
- **Car Accessories** — \$237K (10.90%)
- **Toys** — \$111K (5.09%)
- **Mobile & Accessories** — \$39K (1.80%)

Top Product Performance

Individual product leaders reveal where sales revenue and customer engagement concentrated in 2022.



🏆 Top Sales — Nikon Lens

Wide Angle AF Nikkor 24mm led all products with **\$33,550** in 2022 sales



★ Top Reviews — SanDisk 16GB

3-Pack generated **402,828 reviews** — the highest review volume of any single product

- ❑ The remaining top-5 products by sales and review volume should be populated from the source dataset for a complete ranking.

Key Findings

Four standout insights from the 2022 analysis.



Men Shoes Leads

\$940K — 43.18% of total sales, nearly double the next category



November Peak

\$310K monthly high, driven by pre-holiday demand



Nikon Tops Sales

24mm AF Nikkor lens — \$33,550, highest individual product revenue



SanDisk Tops Reviews

16GB 3-Pack — 402,828 reviews, highest customer engagement

Dashboard Features

Six interactive components support exploration and presentation of Amazon sales data.



→ KPI Cards

YTD sales, QTD sales, products sold, and reviews at a glance

→ Trend Charts

Monthly line and weekly column views of sales movement

→ Category Table

YTD and QTD sales with percentage contribution by category

→ Top-5 Visuals

Leading products ranked by sales and review volume

→ Interactive Slicers

Category and Quarter filters for focused, self-service analysis